

AWS Cloud Security Monitoring

CloudTrail — CloudWatch — S3 — SNS

Project Date: July 30, 2026 | Author: Houston Jones

1. Project Overview

This project demonstrates a complete AWS cloud security monitoring pipeline using native AWS services. The pipeline captures every API call made in the AWS account, stores logs in S3 for long-term retention, streams events to CloudWatch for real-time analysis, and sends email alerts when critical security events are detected — specifically root account usage.

This type of monitoring is a compliance requirement across many regulated industries and directly maps to common security frameworks including SOC 2, PCI DSS, and ISO 27001.

2. Architecture Overview

Service	Role	Purpose
AWS CloudTrail	Log Collection	Captures every API call made in the AWS account across all regions
Amazon S3	Log Storage	Stores CloudTrail logs in compressed JSON format for long-term retention
Amazon CloudWatch Logs	Log Streaming	Receives CloudTrail events in real time for monitoring and alerting
CloudWatch Metric Filter	Detection	Scans log stream for root account usage pattern
CloudWatch Alarm	Alerting	Fires when root account activity is detected
Amazon SNS	Notification	Sends email alert to security team when alarm triggers

3. Configuration Steps

3.1 CloudTrail Setup

- Created trail named: hueylab-cloudtrail
- Enabled multi-region trail — captures events across all AWS regions
- S3 bucket automatically created: aws-cloudtrail-logs-897722702309-9d387704
- Trail status: Logging (active)

- Management events: All API activity captured

3.2 CloudWatch Logs Integration

- Connected CloudTrail to CloudWatch Logs
- Log group created: aws-cloudtrail-logs-897722702309-a033198a
- IAM Role created: CloudTrail-CloudWatchLogs-Role
- 3 log streams active and receiving events in real time

3.3 S3 Log Storage

- CloudTrail logs automatically delivered to S3 every 5 minutes
- Log format: Compressed JSON (.json.gz)
- Folder structure: AWSLogs/AccountID/CloudTrail/Region/Year/Month/Day/
- 52 log files stored on first day of operation
- Storage class: Standard

3.4 IAM User Provisioning — Audit Log Test

Created a test IAM user to generate a provisioning audit event:

- Username: testuser-hueylab
- CloudTrail captured the CreateUser event immediately
- Full audit trail recorded including: who created the user, timestamp, source IP, AWS region, and resource ARN

CloudTrail event details captured:

Field	Value
Event name	CreateUser
Event time	July 30, 2026, 12:03:25 UTC-04:00
User name	root
Source IP address	99.44.50.154
AWS region	us-east-1
Event source	iam.amazonaws.com
Resource created	testuser-hueylab
Read-only	false

3.5 Root Account Usage Detection — Metric Filter

Created a CloudWatch metric filter to detect any root account API activity:

Filter pattern:

```
{ $.userIdentity.type = "Root" && $.userIdentity.invokedBy NOT EXISTS && $.eventType != "AwsServiceEvent" }
```

Metric configuration:

- Filter name: RootAccountUsage
- Metric namespace: CloudTrailMetrics
- Metric name: RootAccountEventCount
- Metric value: 1 (increments by 1 for each root account event)
- Default value: 0 (published when no root activity detected)

Why root account monitoring matters:

The AWS root account has unrestricted access to every service and resource in the account. Best practice is to never use the root account for day-to-day operations. Any root account activity should be treated as a critical security event requiring immediate investigation.

3.6 CloudWatch Alarm Configuration

Setting	Value
Alarm name	RootAccountUsage-Alert
Description	Fires when the AWS root account is used for any API call. Root account usage is a critical security event and should be investigated immediately.
Metric	RootAccountEventCount
Namespace	CloudTrailMetrics
Threshold	RootAccountEventCount >= 1
Period	5 minutes
Datapoints to alarm	1 out of 1
SNS Topic	hueylab-root-account-alert
Notification email	Jjonesholdings93@gmail.com

4. Alert Verification

The alarm was tested by performing actions as the root account user. Within 5 minutes the following occurred:

- CloudTrail captured the root account API call
- CloudWatch Logs received the event from CloudTrail
- Metric filter matched the root account pattern
- RootAccountEventCount metric incremented to 1
- CloudWatch Alarm transitioned from OK to ALARM state
- SNS notification sent email alert at 18:16:39 UTC
- Email received at Jjonesholdings93@gmail.com within 2 minutes

Email alert details:

Field	Value
Subject	ALARM: RootAccountUsage-Alert in US East (N. Virginia)
State Change	OK -> ALARM
Timestamp	Thursday 30 July, 2026 18:16:39 UTC
Reason	Threshold Crossed: 1 out of 1 datapoints >= 1.0
AWS Account	897722702309
Region	us-east-1

5. Security Findings & Recommendations

Finding	Severity	Recommendation
Root account used for IAM user creation	HIGH	Create an IAM admin user for day-to-day operations. Root account should only be used for account-level tasks that require it.
No MFA on root account	CRITICAL	Enable MFA on the root account immediately. Use a hardware MFA device for maximum security.
CloudTrail log file validation disabled	MEDIUM	Enable log file validation to detect if CloudTrail logs have been tampered with.
No S3 bucket encryption	MEDIUM	Enable SSE-KMS encryption on the S3 bucket storing CloudTrail logs.
Log retention set to Never expire	LOW	Set appropriate retention period based on compliance requirements — retention periods vary by industry and regulatory framework.

6. Compliance Relevance

This monitoring pipeline directly addresses requirements from multiple compliance frameworks:

Framework	Requirement Met
SOC 2	CC6.1 — Logical access controls and monitoring of privileged account usage
PCI DSS	Requirement 10 — Track and monitor all access to network resources and cardholder data
SOX	Section 404 — IT general controls including audit logging and change management
AWS CIS Benchmark	CIS 3.3 — Ensure a log metric filter and alarm exist for root account usage

7. Tools & Services Used

Service	Version/Details
AWS CloudTrail	Multi-region trail, management events
Amazon S3	Standard storage class, us-east-1
Amazon CloudWatch Logs	Log group with 3 active streams
Amazon CloudWatch Alarms	Metric alarm with 5-minute period
Amazon SNS	Email notification to Gmail
AWS IAM	Test user creation for audit log generation